

Abdur Razzak

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EDUCATION

- Ph.D. in Computer Science** Aug 2026 – Present
Research Area: High Performance Computing, Modeling and Optimization
University: New Mexico State University, Las Cruces, NM, USA
- M.Sc. in Computer Science** Jan 2023 – Dec 2025
Masters Thesis: Memory Systems Modeling and Prediction
University: New Mexico State University, Las Cruces, NM, USA
Cumulative GPA: 4.00/4.00
- B.Sc. in Computer Science and Engineering** April 2014 – Sept 2018
Undergraduate Thesis: Developing chat-bot ensuring security and privacy issues using Blockchain
University: Shahjalal University of Science & Technology, Sylhet, Bangladesh
Cumulative GPA: 3.44/4.00

TECHNICAL SKILLS

Programming Languages	C/C++, Python, Java, ASSEMBLY
Software Development	Application Life Cycle, Optimization, Integration, Maintenance, Legacy Support, QA and Testing
Version Control	Git, Perforce
Tools and Frameworks	LEX, YACC, Slate, Banner, JUnit, Mockito, Robolectric
Performance Modeling	Clang Compiler, LLVM, Performance Modeling, Static Code Analysis
Parallel Programming	Multi-threading (C, Java), MPI, OpenMP, CUDA
Machine Learning	CNN, LSTM, SVM, YOLO8, YOLO11
Operating System	Linux, Windows, Android
Problem Solving	200+ problems in Data Structures and Algorithms
Programming Competition	Runners up at Fall 2024 and 2023 at Google Developers Student Club Programming Contest at NMSU

WORK EXPERIENCES

- System Developer at The Graduate School (NMSU)** May 2025 – Present
- Graduate Admission System
 - Designing application portals in Slate for the graduate programs with a different set of requirements.
 - Implementing Slate to Banner automation in application data loading.
 - Integrating a secured API system to pull student applications posted on third-party sites.
 - Developing the application review process with multiple stakeholders on campus.
 - CRMs
 - Running a series of automated campaigns for enrollment management, including emails, text, and calls.
 - Designing checkpoints and communication strategies for callers.
 - Incorporating digitalized forms and review process for the current graduate students.
 - Maintaining and implementing new features for the Graduate School website (<https://gradschool.nmsu.edu/>)
- Software Engineer at Samsung R&D Institute, Bangladesh** Sept 2018 – Dec 2022
- Samsung Internet Browser** [Android]
 - Optimized application memory usage by up to 70% by enhancing the Quick Access Module.
 - Implement new features and maintain large codebases.
 - Achieved 95% unit and 85% of User Interface test coverage for Samsung Internet modules.
 - Web Runtime on Samsung Watch** [Tizen]
 - Developed a secured runtime environment on the native C/C++ watch platform for web applications.
 - Improved web application launch time and life cycle performance.
 - Enabled C/C++ projects in cross-platform electron js with legacy web API support. Awarded **ICON of the Month** in 2020.

PUBLICATIONS

- **Abdur Razzak**, Atanu Barai, Nandakishore Santhi, and Abdel Hameed Badawy. 2026. **Compile Time Prediction of Cache Reuse for Array Applications with Multiple Loop Nests**. In Proceedings of the International Symposium on Memory Systems (MEMSYS '25). Association for Computing Machinery, New York, NY, USA, 235–244.
- **Abdur Razzak**, Atanu Barai, Nandakishore Santhi, and Abdel Hameed Badawy. 2024. **Static Reuse Profile Estimation for Array Applications**. In Proceedings of the International Symposium on Memory Systems (MEMSYS '24). Association for Computing Machinery, New York, NY, USA, 235–244.
- Atanu Barai, Nandakishore Santhi, **Abdur Razzak**, Stephan Eidenbenz, and Abdel-Hameed A. Badawy. 2024. **LLVM Static Analysis for Program Characterization and Memory Reuse Profile Estimation**. In MEMSYS '23, ACM, Article 3, 1–6.
- Md Saiful Islam Bhuiyan, **Abdur Razzak**, et al. **BONIK: A Blockchain Empowered Chatbot for Financial Transactions**. In TrustCom 2020, IEEE, pp. 1079-1088.

PROJECTS

- System Programming and Simulation Tools
- **C-Compiler**: Built a compiler that takes C programs as an input, generates the symbolic table, creates the Abstract Syntax Tree (AST) and outputs machine readable assembly code. **LEX, YACC, C, and Assembly**. Spring 2024
 - **Cache Simulator**: Created a cache simulator tool (L1 cache) to evaluate cache hit rate for given memory traces with customizable settings for cache size, associativity, and write policies. **C**. Spring 2023
- Parallel Programming
- **Evaluate Platforms**: Analyzed runtime performance of five parallel platforms with datasets ranging from 10K to 1M. **C++, Java, MPI, OpenMP, CUDA** Spring 2024
 - **Optimizing Thread-Level Parallelism**: Improved synchronization overhead and cache efficiency considering memory coherence and consistency for multi-threaded programs. **C++, Multi-threading** Spring 2024
- Machine Learning
- **Multi-modal vs Single-Entity Model Comparison**: Developed a model that learns from both images and text using a combination of Convolution Neural Network (CNN) and Long Short-Term Memory (LSTM). **Python, SVM, CNN, LSTM** Fall 2023
 - **Signature Verification**: Built a CNN (Convolution Neural Network) model that detects genuine or fraudulent signatures with 92% accuracy. **Python, TensorFlow, CNN** Fall 2018
- Web Development
- **Checkers 2-Player Game**: Developed an interactive online game where players move pieces across a board. **JavaScript, Socket Programming, HTML, CSS** Fall 2023
 - **Selenium Browser Task Automation**: Automated application processing tool to complete repetitive tasks on a browser tab using selenium. **Python, Selenium, HTML** Spring 2024

RELEVANT COURSEWORKS

Parallel Programming Compiler & Automata	Computer Architecture(I & II) Data Mining	Data Structures & Algorithm Transitions Advanced Machine Learning
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REFERENCE

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